# Symphony: A Heuristic Normalized Calibrated Advantage Actor and Critic Algorithm with Application to Humanoid Robots


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***Abstract*—In our work we implicitly suggest that it is a misconception to think that humans learn fast. The learning process takes time. Babies start learning to move in the restricted fluid environment of the womb. Children are often limited by an underdeveloped body. Even adults are not allowed to participate in complex competitions right away. However, with robots, when learning from scratch, we often don't have the privilege of waiting for tens of millions of steps. "Swaddling" regularization is responsible for restraining an agent in rapid but unstable development penalizing action strength in a specific way not affecting actions directly.**

**The Symphony, Transitional-policy[1] Deterministic Actor and Critic algorithm, is a concise combination of different ideas for possibility of training humanoid robots from scratch with Sample Efficiency, Sample Proximity and Safety of Actions in mind. It is well known that a continuous increase in Gaussian noise without appropriate smoothing is harmful for motors and gearboxes. Moreover, an added noise trail, when actions are clipped, can accumulate at the action limits, potentially increasing the frequency of maximum control values (torque, velocity, angles). Compared to Stochastic algorithms, we use a different approach, we set a limited parametric noise and adjust the strength of actions instead of noise. Promoting a reduced strength of actions, we safely increase entropy, since the actions are submerged in weaker noise. When actions require more extreme values, actions rise above the weak noise and become more deterministic. Training becomes empirically much safer for both the surrounding environment and the robot's mechanisms.**

**We use Fading Replay Buffer: using a fixed formula containing the hyperbolic tangent, we adjust the batch sampling probability: the memory contains a recent memory and a long-term memory trail. Fading Replay Buffer with increased Update-To-Data ratio (3) allows us to use Temporal Advantage when we improve the current Critic Network prediction compared to the exponential moving average. Temporal Advantage allows us to update the Actor and the Critic in one pass, as well as combine the Actor and the Critic in one Object (Class) and implement their Losses in one line. The Symphony algorithm is called Heuristic[2] and Calibrated, since the activation, loss, regularization functions were rewritten and parameters were customized during numerous experiments in order to bring safe model-free imitation-less Humanoid robot training closer to reality.**


[1]algorithm in between on and off policy due to balanced updates

[2]a hypothetical approach of combining different methods and using specific parameters based on intuition and experience to solve practical problems


***Index Terms*—sample-efficient, model-free, off-policy, advantage, reinforcement learning**


## I. PRELIMINARY

We consider an Agent or Actor that utilizes a Policy $A_\phi$ or Actor Network with parameters $\phi$ which at a Continuous Markov Decision Process (MDP) State (hereinafter referred to as simply "state") $s_t$ produces a vector of continuous actions (hereinafter "actions") $a_t + \epsilon \in [-a_{max}, a_{max}]$. The Environment responds with reward $r_t$ and the Agent appears at the Next MDP State (next state) $s_{t+1}$. We use the terms Critic, Critic Network, and Q Network interchangeably, while the Critic's parameters are abbreviated by $\theta$. Critic and Target Critic output $q$ and $q^{'}$ value predictions, respectively, and * abbreviation is used for values detached from the computation graph to compute gradients.

## II. INTRODUCTION

The Symphony algorithm aims to solve two conflicting problems of Model-Free Imitation-Less Reinforcement Learning: Sample Efficiency and harmonious agent motion without jerky movements, which is achieved with the help of Sample Proximity and Safety of Actions. On-policy algorithms like Proximal Policy Optimization [1] (PPO) solved the problem of harmonious agent movements using a small "safe" gradient step, which leads to a larger number of close samples and better policy consequently. However, PPO has proven effective in simulations, where the learning process can be parallelized across many virtual environments using Graphics Processing Units or Tensor Processing Units, and make a gradient step based on a larger number of roll-outs, so this process requires a careful transition from simulation to a real robot, though no bootstrapping (updating Online Q Network based on prediction of its delayed variant) is used. When it comes to training on a real robot from scratch, entropy regularization based off-policy sample-efficient algorithms like Soft Actor- Critic [2] (SAC) and its advanced derivatives are often used, though this family of algorithms can suffer from prediction inaccuracies of the TD equation; the regularization mitigates this issue to some degree.

To achieve even higher Sample-Efficiency of the SAC algorithm, recent studies (SAC-20, Randomized Ensembled Double Q learning or RedQ [3]) have concluded that it is possible to do a larger number of updates-per-frame. Though taking the minimum of two prediction functions of the future aggregated reward min(Q1, Q2) [4] at an increased update frequency increases the probability of stagnation of the policy or Actor Critic at a local extremum due to the constantly decreasing step of min(Q1, Q2); Random Q network selection from Q network ensemble (RedQ algorithm) helps against this. The Aggressive Q learning with Ensembles [5] (AQE) algorithm instead of min(Q1, Q2) uses the idea of Truncated Quantile Critics [6] (TQC) which is about taking the average in a larger number of Q heads or nodes (Q distribution) after truncating several highest Q nodes. Truncation is necessary due to the overestimation and high variability of the average value (for Walker-2d this is up to half of the nodes). This step allows achieving highest scores. Another work (Reset [7] algorithm) suggests leaving the minimization but refreshing the weights of the last two layers to the default ones, periodic refresh allows using a higher update frequency, up to 128.

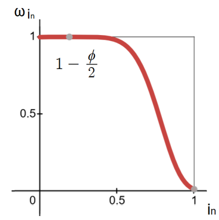
If we look at the work of Distributional Soft Actor and Critic [8], we can understand that the average value is a kind of ”Pandora’s box” for humanoid agent movements by increasing the variation and the number of attempts or roll-outs [9]. The point is that when we take the minimum of two functions for predicting the best actions, with time we create a very careful, cautious or risk-averse agent, but when we take the average, we give the agent the opportunity to be relatively optimistic in its predictions and make more mistakes; for humanoid agents, this means more falls. But when the agent falls more, this in turn leads to better experience, which results in more harmonious movements and more deterministic $Q = r_{done}$ in the batch sample, which in the Temporal Difference equation provides better predictions in the end. Unfortunately, this takes some time, while introducing a high instability in the learning process - the first version of Distributional Soft Actor and Critic still required several millions of steps. Most likely, because of this, the authors of this algorithm returned to the minimum between two Q distributions [10]

## III. Foundation

The Symphony is an advantage based transitional-policy deterministic Actor-Critic algorithm with limited[3] Normal Gaussian noise ($\epsilon \sim a_{max} * 1/e * \mathcal{N}(0.0, 1.0)$, where $\mathcal{N} \in [-e, e]$) added to output actions $a$.

In Symphony, instead of ensemble of 5-10 networks or only 1-2 Critics, we take 3 Critics each of which outputs $n_q$ (128) nodes and concatenate output nodes together producing a distribution of size $N_q$ (384). For the output of the Target Critic, we sort the distribution from minimum to maximum and multiply those values by fixed weights that are opposite to the Fading Replay Buffer probabilities (which will be discussed further) though for the length of $N_q$. The weighted sum of $N_q$ Critic’s nodes becomes the output of the Target Critic (higher q values are smoothly truncated).

TABLE I: Fixed weights for Target Critic’s nodes




1) $i_n = \frac{\{1,2,3 \dots N_q\}}{N_q}$

2) $w_{i_n} = e^{-|\frac{1-b-i_n}{s}|^{2e}}$

3) $w_n = w_{i_n} / \sum_1^{N_q} w_{i_n}$

$$q = \sum_{n=1}^{N_q} (w_n * q_n)$$

$b = \phi/2$, $s = 1/\phi$, $\phi = (\sqrt{5}+1)/2 \approx 1.618$

Symphony increases Temporal Advantage (TA) or plain difference between a currently predicted target q value and exponential weighted moving average of target q values (copy of q values, $q^*$, detached from the gradient computation) with heuristic $\alpha = 1/\phi = \frac{\sqrt{5}-1}{2} \approx 0.618$ (a reciprocal of golden ratio)

The Symphony’s Objective Function can be expressed as ($Q_T = \alpha \, \bar{Q}_{T-1} + (1-\alpha) \, q^{'*}_{t+1}$, where $q^{'*}_{t+1}$ is the next target Q Network prediction detached from gradient computation):

$$\mathbb{E}_{s_t \sim \rho^\beta} [\nabla_a \left[ Q_{target}(s_{t+1}, \tilde{a}(s_{t+1})) - Q_T \right] \nabla_\phi A(s_{t+1}|\phi)] \tag{1}$$

Though, Target Critic based Actor’s updates can slow down training due to Polyak Averaging [11] [12] via soft updates ($\theta^{'} \leftarrow \tau\theta + (1-\tau)\theta^{'}$) [13], Advantage or $\Delta Q$ gives us some independence from the actual q values. This method also provides a ”seamless” computationally efficient update of the Actor-Critic. For this we predict the next action $a^{'}_{t+1}$ using the next state and Online Actor ($\tilde{a}_{t+1} = A_\phi(s_{t+1})$), then use the next state and the next action to produce the next target $q$ value:

$$q^{'}_{t+1} = Q_{target}(s_{t+1}, \tilde{a}_{t+1}) \tag{2}$$

which is used to update simultaneously Online Actor via Temporal Advantage and its detached variant is used to update Online Critic via Temporal Difference [14]:

$$-L_\phi\left(\frac{q^{'}_{t+1} - Q_T}{|Q_T|}\right) + L_\theta(\hat{r}_t + \gamma q^{'*}_{t+1} - Q_{online}) \tag{3}$$

Simultaneous update of Online Networks becomes possible when we combine the Actor and Critic into a single Class, or unified Object consequently. We do not utilize Target Actor, while the Target Critic is updated using Polyak Averaging as previously mentioned.

[3]The parametric standard deviation of $1/e$ scales Standard Gaussian noise’s range to [-1, 1]

*Equation revision ...*

One can notice $|Q_T|$ in the denominator of TA, which is used for Advantage normalization, while $\widehat{r_t}$ are normalized rewards. Normalization is performed using $\bar{r}_n$, a mean of absolute values of the exploratory rewards $r_{exp}$ collected during $N^{exp} = 10,240$ exploration steps:

$$\bar{r}_n = \frac{1}{N_{exp}} \sum_{1}^{N_{exp}} |r_{exp}| \tag{4}$$

Normalization is done after exploration and further using estimated $\bar{r}_n$.

*... end of revision*

As the Replay Buffer grows in size, sampling evenly with a small batch size (e.g., 32–64) may lead to Q-values predicted from one batch having a very weak correlation with Q predictions from the other. To address this, we use the previously mentioned exponential smoothing, an increased batch size of 384, and a Fading Replay Buffer, where sampling priorities $p$ correspond to the normalized indexes of the transitions $i$ (normalized by the capacity of the Replay Buffer, $N_{rb}$).

We then used a bell-shaped Gaussian with a power of $2e$, a width of $1/\phi$, and a shift of $0.5 + 0.5\,(1/\phi) = \phi/2$, which has a maximum at the shift point $\phi/2$. This created a plateau like shape for the latest experience and a longer fading trail.

To obtain probabilities that add up to 1.0 we further divide weights/priorities by a sum of all weights/priorities.

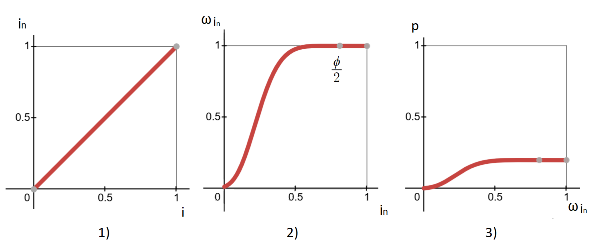
TABLE II: Probabilities for the Fading Replay Buffer

1) $i_n = \frac{\{1,2,3 \ldots N_{rb}\}}{N_{rb}}$

2) $w_{i_n} = e^{-|\frac{i_n - b}{s}|^{2e}}$

3) $p = \frac{w_{i_n}}{\sum_1^{N_{rb}} w_{i_n}}$

$b = \phi/2$, $s = 1/\phi$, $\phi = (\sqrt{5}+1)/2 \approx 1.618$




Finally, we repeat 20,000 Exploration transitions 50 times, creating a fully filled Replay Buffer from the beginning. In this way we ensure that all Exploration transitions will be thoroughly processed before disappearing. After the Replay Buffer is filled, a new transition enters it at the last index. Then we do roll or shift left operation so that the first (oldest) transition appears at the last index and is ready to be overwritten. When we encounter a transition containing $\widehat{r_t} = r_{done}$ at the oldest placeholder, we shift left by two positions instead of one and subtract a small value, $\epsilon$, from the 'done=1.0' indicator. This ensures that terminal transitions are processed for one extra cycle before being overwritten. This resembles human memory which remembers pivotal points and makes the Actor-Critic Network less dependent on Bootstrapping.

Furthermore, outside of the Replay Buffer, when agent learns to move without falling or approaches an artificial termination limit, e.g. 1000, at a timestep close to termination (950) we zero the actions $a_t$ and leave only noise, so that the agent falls deliberately.

The Fading Replay Buffer is an intermediate step between On-policy Cache type Memory and Off-policy Experience Replay [15]. It smooths difference between both memories since actions that were done far in the past have a little relevance to the current policy, but it is also important to improve Q from those old "bad" transitions for better generalization. Also Fading Replay Buffer diminishes a transition problem of Experience Replay as a vanilla Experience Replay is a Stack type memory until it reaches its capacity then it is a Double-ended queue or a Standard Moving Average.

Temporal Advantage and Fading Replay Buffer introduce high sample efficiency which can lead to the fast growth in Q learning. However, if more closely related experiments lead to more harmonious agent movements but requires large number of samples, then how to achieve Sample Proximity and Sample Efficiency if these two goals contradict each other?

It may seem counterintuitive but encouraging the agent not to use the full range of motion, "Swaddling" or Control Cost punishment, can help.

Before we learned to run, we learned to walk, before we learned to walk, we learned to stand. If a small child is not swaddled and not kept in a crib, he can cause a lot of trouble to himself and to the Environment. How is swaddling different from "caution" in predictions? In that variability is preserved, but in a small range. This means that before learning to run like Usain Bolt, our agent must learn to move its legs correctly.

If we look at the reward formation in the Mujoco Humanoid-v4 environment, we subtract the sum of squares of actions from the positive direction of speed. That is, we punish the agent for the power or strength of actions – the stronger the actions, the stronger the punishment, although the weight of this punishment (control cost) is 0.1, and the weight of the speed reward is 1.25, and the healthy state is 5. It turns out that the relative "Swaddling" is embedded in the reward. But the weight value for the control cost in Walker-2d is only 0.001. When we do a comparative analysis using these environments, the environments themselves act as a black box, and we cannot influence the internal parameters of the environment (which does not apply to developing our own robot) What is "Swaddling" regarding the algorithm? In the simplest implementation, it is a direct soft suppression of bigger actions by adding control cost to the Actor's Loss Function:

$$ctrl\_cost = \beta\, a^2 \tag{5}$$

where $\beta$ is a temperature parameter, a - actions. $a^2$ in Loss function will reduce the strength of the actions depending on the $\beta$ coefficient. The $\beta$'s strength should be relatively small, otherwise we are unlikely to learn deterministic moves. Adjusting $\beta$, we facilitate actions that can be viewed as buried in limited Gaussian Noise increasing Entropy Level. The difference from a learnable standard deviation is that Noise Level is set constant, it will never go to unprecedented values.

With the introduction of a new harmonic activation function that naturally induces fluctuations in the output and with new loss functions that allow for a less strict penalization of stronger values, and using the principles behind these functions, instead of adding the control cost, we introduced a new approach called "Swaddling" which does not affect actions directly.

*Rectified Sine activation function*

Harmonic activation functions are rarely used in Reinforcement Learning due to high levels of instability. But if stabilized they can introduce infinite exploration in a more natural way than hard resetting neural networks introduced in Reset mechanism [7] as small input signal can cause some oscillatory effect (as in the muscular system of animals).

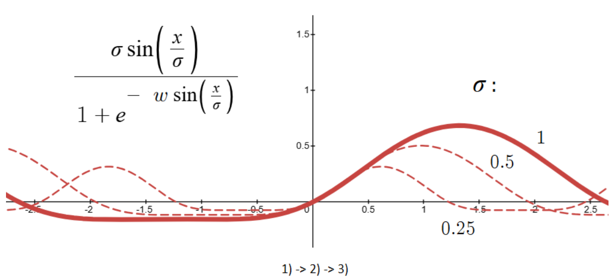
We use internal Sine wave function in between two Linear Layers both in Actor and Critic, which represents an approximation to a Fourier Series in the form of: $Asin(wx+b)+c$, where an absence of the Cosine part is to some extent covered by the presence of a learnable phase shift. A harmonic and non-linear nature of a Sine wave introduces instability and consequently a broader search for solutions in the learning process. We rectified those instabilities by a Swish or SiLU activation function with a weight factor $w$.

TABLE III: Rectified Sine or ReSine

$$1)\ \sigma = \text{sigmoid}(s)\ ;\ w = \frac{1}{\text{sigmoid}(b)}$$

$$2)\ f(x) = \sigma \sin\left(\frac{x}{\sigma}\right)$$

$$3)\ F(x) = f(x)\ \text{sigmoid}\left(w\,\frac{f(x)}{\sigma}\right)$$




During experiments, we slightly decreased the scale of the Rectified Sine function, which showed better balance between exploration and exploitation. However, the best choice was to implement proportional scaling ($\sigma$) using a trainable vector s with a length equal to the hidden dimension (initialized uniformly between -$\frac{1}{\sqrt{dim}}$ and $\frac{1}{\sqrt{dim}}$). After this vector is squashed between 0 and 1 using a sigmoid function, it is then used to proportionally scale the activation function.

Instead of using a fixed value for $w$, we substitute it with the reciprocal of a trainable vector $b$ passed through a sigmoid function, similar to $s$. Consequently, as the denominator varies from 0 to 1, the function transitions between being rigidly rectified and more sinusoidal. The vector $b$ is initialized using a uniform distribution within the range of $\left(-\frac{1}{\sqrt{dim}}, \frac{1}{\sqrt{dim}}\right)$.

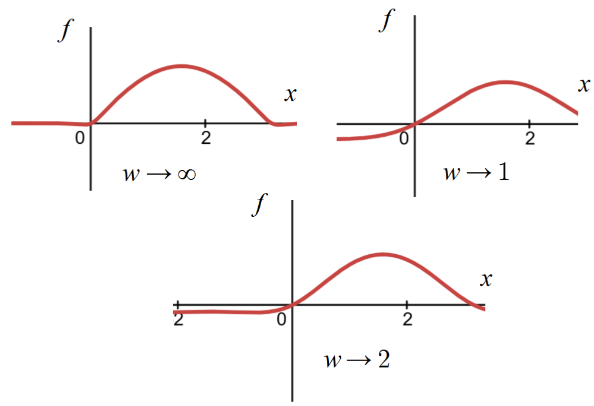



Fig. 1: An effect of different $w$ values

Apart from stabilization, rectification additionally provides pseudo-random harmonic dropouts at rectified areas, while noise produced in the form of harmonics is visually closer to the movements of living beings.

*Rectified Huber Symmetric and Asymmetric Loss Functions*

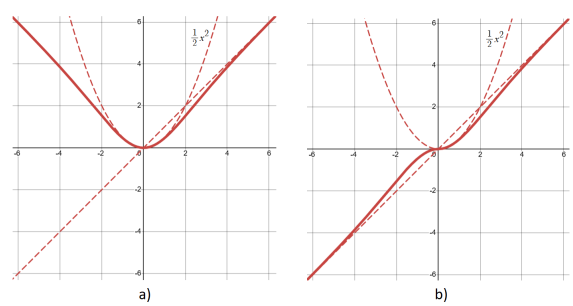



Fig. 2: a) x*tanh(x/2) - Rectified Huber Symmetric Error, b) abs(x)*tanh(x/2) - Rectified Huber Asymmetric Error.

We discussed Rectified Huber Symmetric $x\,tanh(x)$ and Asymmetric $|x|\,tanh(x)$ Error (ReHSE, ReHAE) functions previously [16] [17]. x is the error between prediction and target for the Loss function (ReHSE) or prediction and baseline for the Advantage function (ReHAE).

They have almost quadratic relationships $x^2$ for relatively small values, but the functions strive for linearity for big values as tanh approaches 1 for bigger values. The difference with the Huber loss function is that transition happens gradually. Symmetric version ($x\,tanh(x)$) is used to increase stability in

Critic learning, while faster learning can be achieved when the Asymmetric version ($|x|\, tanh(x)$) is used in conjunction with Advantage for Actor's updates. For the Advantage it is important to keep the sign intact.

Delta or Advantage utilizes gradual change from quadratic to linear relationship most of the time, whereas Q value prediction even if it starts from low values initially, rapidly goes to stronger values ($\gg 1.0$) for positive rewards. The benefit of quadratic relationship at smaller values is a dampening effect of small deterministic gradients. Additionally, we made ReHSE and ReHAE closer to the Huber representation by dividing the argument under tanh by 2: $xtanh(x/2)$ and $|x|tanh(x/2)$.

The expression $tanh(x/2)$ can also be used for a sparser squashing of actions before adding noise. And when noise is introduced, a more linear squashing using $tanh(x)$ instead of action clipping can help prevent noise accumulation at the action limits. At the action edges, Gaussian noise will become slightly skewed toward the center with a narrower distribution due to the applied tanh function. Considering action regularization, both processes can be combined into the representation shown in Figure 3, where $N = \frac{1}{e}\,\mathcal{N}(0.0,\, 1.0)$, $\mathcal{N} \in [-e, e]$.

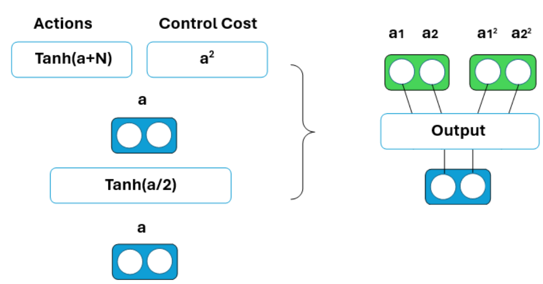



Fig. 3: "Control Cost" regularization.

Unfortunately, the direct relationship between "Control Cost" regularization and actions can impose stronger constraint to the learning process, thus the temperature $\beta$ should be infinitely small.

*Decoupling action regularization from actions*

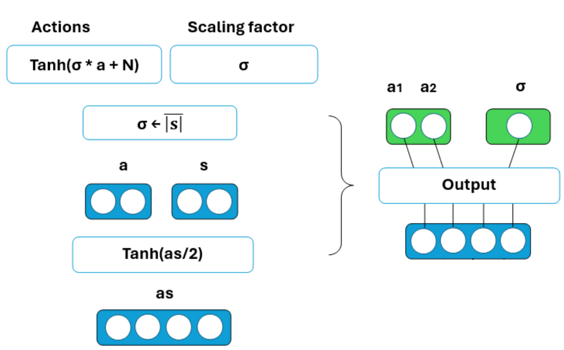



Fig. 4: Action decoupling process

We can use a relatively similar approach that we used in proportional scaling for the ReSine activation function. We can limit the action scaling using the limiting vector s. In this case, the Actor Network will output two independent parameters a and s, both with the action dimension. We can process s through the same $tanh(x/2)$ function. The only thing we need to do is to find mean over absolute values of vector s (the average counteracts overfitting), Figure 4.

For regularization of $\sigma$, we can use inverse tanh (ATanh) with argument $\sigma^{1/\beta}$ creating a barrier regularization for values approaching 1.0. Under regularization ATanh strives for its minimum value of 0.0, while $\beta$ responds for creation of a symmetric plateau around 0.0. To differentiate it from the control cost ($a^2$) we referred to it as "Swaddling" ($\Omega$)

We must multiply output actions by $a_{max}$, for simplicity we omitted this step in the figure. A factor 2 is used for Swaddling, as $2\, ATanh(x)$ is inverse for $Tanh(x/2)$.

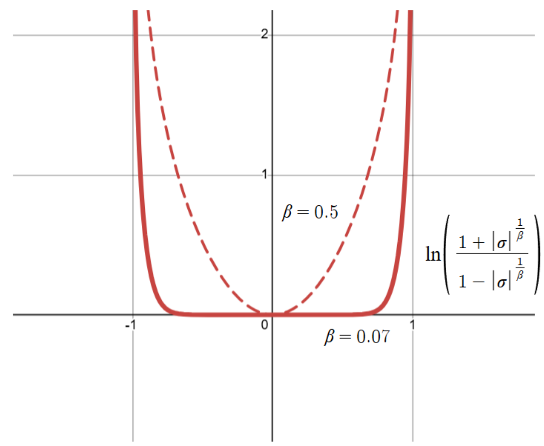



Fig. 5: Swaddling Regularization $\Omega$ with $\beta$ as temperature.

*Adjusting $\beta$*

When the degree $\beta = 0.5$, the function $\Omega(x) = \ln\left(\frac{1+x}{1-x}\right)$ (where $x \leftarrow |\sigma|^{1/\beta}$) resembles a hyperbola near 0.0, while it approaches +inf as $\sigma$ approaches 1.0. When we decrease $\beta$ we give a broader range for our scaling factor to operate with, e.g. $\beta = 0.07$.

To increase entropy, the $\beta$ value should suppress scaling factor values above the standard deviation of noise, but prevent the scaling factor from rolling toward 0.0. To determine $\beta$ more precisely, we can use a nontrivial approach by adding a helper function $\omega(x) = x\, ln(x)$. This function's minimum is reached at $x = \frac{1}{e}$, precisely the value of the parametric standard deviation. However, if we add this function with the $\beta$ parameter to align it with ATanh, then for a strong beta, the minimum will shift closer to the center, while we want the scaling factor to float around the standard deviation. The combined equation $\Omega_\omega(\sigma, \beta) = ln(\frac{1+\sigma^{1/\beta}}{1-\sigma^{1/\beta}}) + \beta\sigma ln(\sigma)$ (where $\sigma, \beta > 0$) is difficult to solve mathematically for $\beta$, but heuristically, we see that values $\beta \leq 0.05$ provide the correct match (for values smaller than 0.05, the scale value for the function's extremum coincides with 1/e):

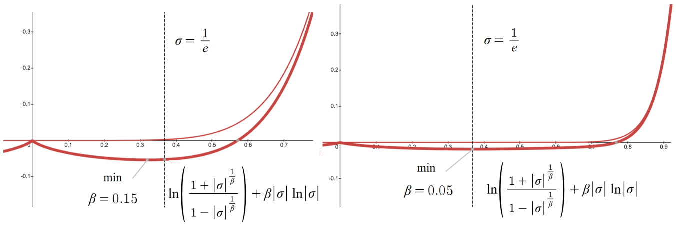


Fig. 6: Approximating $\beta$ value

For adjustable Beta, it is possible to make it trainable. We repeated the previous approach where the Actor Network outputs several parameters, now $\sigma$ and $\beta$ additionally to actions, except we do not find means of absolute values of vectors s and b because $\sigma$ and $\beta$ are to balance each other to some extent, though they are clipped between $1\mathrm{e}{-3}$ and $1.0 - 1\mathrm{e}{-3}$:

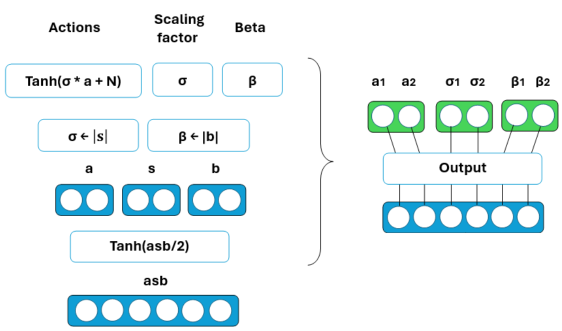



Fig. 7: $\beta$ as trainable parameter

$2\,ATanh(x)$ is also used to regularize Beta, with the function argument being $\beta^{1/0.5}$ or $\beta^2$. A separate helper function for Beta is not needed, since $\beta$, by the defined property, should converge closer to 0.0. While scaling factor $\sigma$ participates in Q value learning through produced actions, $\beta$ mostly is a resisting factor. Sending $\beta$ without power of 2 will result in a higher gradient value directed towards 0.0 which quickly diminishes $\beta$; while sending $\beta$ with a higher order in opposite decreases the gradient and imposes a stronger obstacle for Q value learning through increased $\beta$. On the one hand the helper function already steers Beta toward 1.0 (function's minimum), on the other the added hyperbola shaped regularization suppresses its value ($\beta^*$ is a detached value from the gradient computation):

Similar to the log probability used in the SAC framework, we calculate a sum across the $\sigma/\beta$ dimensions (equal to the action space) for Swaddling Regularization and for the $\beta$ loss instead of computing the average. This variant is called Symphony-S2. While the average is still possible to decrease an entropy and achieve top scores faster, but by reducing entropy, we increase Sample Efficiency at the expense of wider action scale and worse generalization, which impacts Safety of Actions and Sample Proximity, respectively. We designate this version as Symphony-S.

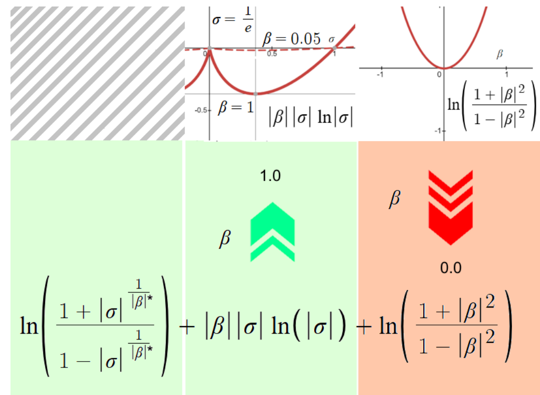



Fig. 8: Swaddling Regularization $\Omega_\omega$ and $\beta$ loss.

*Control Cost and Critic Regularization*

Swaddling regularization is responsible for directly regularizing the Actor Network (Action Safety), whereas the environment's intrinsic control cost can help generalize the Critic Network, thereby passively regularizing the Actor for long-horizon actions (Action Proximity). We modified the control cost, decoupling it from the environment's reward function and introducing penalty into the TD Bellman Equation by subtracting a Swaddling regularization's value from the next Q prediction. By doing this, we are rewarding action scales that are close to $1/e$ while severely penalizing proximity to the maximum scale boundaries:

$$q_t^{'*} = \widehat{r}_t + \gamma \left( q_{t+1}^{'*} - \mathcal{H}_{t+1}^{*} \right) \tag{6}$$

where $\mathcal{H}_{t+1} = \Omega\omega(\sigma_{t+1}, \beta_{t+1})$. Due to the seamless unified update, we do not recalculate the Swaddling Regularization's value.

*AdamW Optimizer with diminishing $\epsilon$*

The Actor-Critic network's Optimizer is AdamW [18] [19] with the learning rate $\alpha_{lr} = 1\mathrm{e}{-4}$, less sharp gradient with $\beta 1, \beta 2 = (\alpha, 1-\tau) \approx (0.62, 0.995)$ and weight decay $\lambda = 0.01$. Performance optimization: we removed Adam's [18] default bias correction as it is not essential in the Reinforcement Learning setup (the native bias correction becomes negligible after approximately 1,000 training steps.); the weight decay complements the main gradient value without a significant alteration of AdamW's principles:

$$\theta_t \leftarrow \theta_{t-1} - \alpha_{lr} \left( m_t/(v_t + \epsilon) + \lambda\, \theta_{t-1} \right) \tag{7}$$

Due to the convergence of the numerator and denominator values inherent in RMS-prop type optimizers, the gradient update step in the Adam optimizer rapidly approaches 1, ultimately resulting in unnatural agent poses that require prolonged recovery phases.

To address the update explosion, we propose scheduled damping of the $\epsilon$ (epsilon) parameter in the Adam denominator. By annealing $\epsilon$ from a value close to 1 down to a value close to 0 at a rate governed by $\beta_2$, we obtain an optimizer that smoothly transitions from SGD to Adam. This transition provides sufficient time for the stabilizing normalization parameter in the denominator to accumulate. The scheduling formula is defined as follows ($\epsilon_0 = 1-10^{-8}$,):

$$\epsilon_t = \epsilon_{t-1} \cdot \beta_2 + 10^{-8} \cdot (1 - \beta_2) \tag{8}$$

which is used in:

$$\theta_t \leftarrow \theta_{t-1}\lambda^* - \alpha_{lr}\frac{m_t}{v_t + \epsilon_t} \tag{9}$$

where $\lambda^* = 1 - \alpha_{lr}\lambda$ is a parametric reduction.

*Gradient Dropout*

We use Layer Normalization [20] after the first fully connected layer. Layer Normalization and Resine Activation Function provide indirect generalization for the Input and Hidden Layers, respectively, whereas the Latent Layer was left untouched. Original Dropout Layer [21] did not suit as it would zero some nodes distorting output distribution.

For the latent layers we added a specific implementation of *Gradient Dropout* [22], which instead of zeroing some nodes as in Dropout Q Functions (DroQ) [23], does not calculate their gradients during backpropagation. This can be achieved simply by (mask is an original dropout mask, $x^*$ - detached values):

$$mask * x + (1 - mask) * x^* \tag{10}$$

Gradient Dropout after Fully Connected Output Layer in Actor and Critic with a dropout probability of $50\%$ was the last missing ingredient. This choice followed extensive experimentation with different dropout rates, where $p = 0.5$ was chosen as a robust baseline since no single value proved superior. Motivated by these findings, we transitioned to a stochastic approach, passing a standard Gaussian distribution through a sigmoid function to determine the probability.

$$p_t = \sigma(x), \quad \text{where } x \sim \mathcal{N}(0, 1) \tag{11}$$

Although the dropout probability may fluctuate or show some bias early on, it averages out to exactly 0.5 over a large number of training iterations.

*Hyper-parameters matter*

In Symphony algorithm we rigidly preset some constants that can also effect the final performance. Reward normalization factor $\bar{r}_n = \overline{|r_{exp}|}$ calculated after $N_{exp} = 20,000$ exploratory steps, which are repeated 50 times to produce a fully filled memory from the beginning. The Fading Replay Buffer's capacity of $N_{rb} = 50\,N_{exp}$, decay factor $\gamma = 0.99$, delay constant $\tau = 0.001$, Update-To-Data $\mathcal{G} = 1$.

The reciprocal of the golden ratio for the smoothing factor of the TA's exponential moving average $\alpha = \frac{1}{\phi} = \frac{\sqrt{5}-1}{2} \approx 0.62$.

The batch-size $\mathcal{B} = 384$. The hidden layer size $h_{dim}$ is 768 for Actor-Critic inner networks. Each Critic Network has $\mathcal{B} \div 3 = 128$ output nodes, so that the number of concatenated output nodes of the Critic Network coincides with the batch-size $\mathcal{B}$.

Each step during and after Exploration we set new seeds for the libraries we work with (namely, PyTorch, numpy, random).

After the Replay Buffer is filled with the Exploratory data, a new transition enters it at the last index. Then we do roll or shift left operation so that the first (oldest) transition appears at the last index and is ready to be overwritten. When we encounter a transition that contains $\widehat{r_t} = r_{done}$ at the oldest placeholder, we do shift left 2 positions instead of 1. At a step close to the artificial termination limit we zero scaled actions and leave only noise.

An important adjustment to increase speed of training is to place the Replay Buffer on the same device the network's learning is accomplished: we created State, Action, Reward, Done, Next State placeholders with size ($N_{max}$, feature dimension). It is significantly more computationally efficient to put single instance of each feature to the device placeholder than to sample a bigger batch and put it from one device to another at each training step, not to mention that instances can be accessed just by list of indexes.

We additionally optimized the Fading Replay Buffer by refactoring the resource-consuming roll operation into a circular buffer with a cyclic pointer, while also implementing in-place operations to reduce memory overhead.

To decrease computational time further we recommend performing a PyTorch's trivial just-in-time (JIT) compilation on new modules converting them to C++ optimized graphs including Loss Functions which are processed as modules with forward pass.

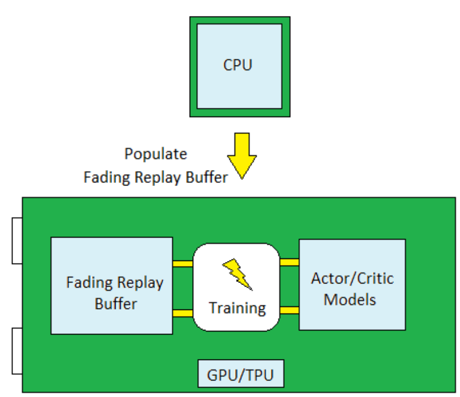



Fig. 9: Representation of Device occupation and training

**Symphony: Pseudo-code**

Initialize Actor-Critic Network ($A_\phi$, $Q_\theta$);
Initialize Target Critic Network $Q_{\theta'}$ ($\theta' \leftarrow \theta$);
Initialize Fading Replay Buffer $\mathcal{F}(N_{rb})$;
Set default $\alpha_{lr}, \gamma, \tau, \mathcal{B}, \mathcal{G}$. Set $\alpha = \frac{1}{\phi}$;
After $N_{exp}$ estimate $\bar{r}_n$ and normalize $r_{exp}$;
Fill $\mathcal{F}$ by repeating $N_{exp}$ transitions 50 times, update priorities in $\mathcal{F}$;
**foreach** *episode* **do**
  **foreach** *step of episode* **do**
    Take an action $a_t$, observe $r_t$, $s_{t+1}$;
    Store $i$ transition $(s_t, a_t, r_t, s_{t+1})$ in $\mathcal{F}$;
    Sample $\mathcal{B}$ $(s_t, a_t, r_t, s_{t+1})$ transitions using priorities from $\mathcal{F}$;
    **foreach** $\mathcal{G}$ *updates* **do**

$$\tilde{a}_{t+1}, \sigma_{t+1}, \beta_{t+1} = A_\phi(s_{t+1})$$

$$q'_{t+1} = Q_{\theta'}(s_{t+1}, \tilde{a}_{t+1})$$

$$\mathcal{H}_{t+1} = \Omega_\omega(\sigma_{t+1}, \beta_{t+1})$$

$$q'^{*}_{t} = \widehat{r}_t + \gamma\left(q'^{*}_{t+1} - \mathcal{H}^{*}_{t+1}\right)$$

$$Q_T = \alpha\,\bar{Q}_{T-1} + (1-\alpha)\,q'^{*}_{t+1}$$

$$L_{\phi,\theta} = -\mathrm{ReHAE}\left(\frac{q'_{t+1} - Q_T}{|Q_T|}\right) +$$

$$+\,\mathrm{ReHSE}(q'^{*}_t - Q_\theta(s,a)) +$$

$$+\,\mathcal{H}_{t+1} + \Omega(\beta^2_{t+1})$$

$$\phi \leftarrow \phi - \alpha_{lr}\frac{\partial \bar{L}_{\phi,\theta}}{\partial\,\phi}$$

$$\theta \leftarrow \theta - \alpha_{lr}\frac{\partial \bar{L}_{\phi,\theta}}{\partial\,\theta}$$

$$\theta' \leftarrow (1-\tau)\theta' + \tau\theta$$

    **end foreach**
    Change random seeds;
  **end foreach**
**end foreach**

---

| | | | |
|---|---|---|---|
| $*$ : | values detached from gradient computation | | |
| $\tilde{a}$ : | predicted actions | $'$ : | target values |
| $\bar{r}_n$ : | $\overline{\lvert r_{exp} \rvert}$ | $\widehat{r}_t$ : | $r_t/\bar{r}_n$ |
| ReHSE(x): | $x\,tanh(x/2)$ | ReHAE(x): | $\lvert x\rvert\,tanh(x/2)$ |
| $\Omega(x)$ : | $ln(\frac{1+x}{1-x})$ | $\omega(x)$ : | $x\,ln(x)$ |
| $\Omega_\omega(x,\kappa)$ : | $\Omega\,(x^{1/\kappa^*}) + \kappa\,\omega(x)$ | | |

Under ReHSE, ReHAE, $\Omega\omega$, $\Omega$ functions mean is calculated over batch of size $\mathcal{B}$. Standard Gradient Descent used as a shorthand for customized AdamW optimizer;

*Partially Observable Markov Decision Process (POMDP)*

For POMDP cases where not all state data is accessible for decision-making, we developed a shared Feature Extractor for the Actor, Critic, and an additional Reward Network, building upon the widely known Online Feature Extractor (OFE).

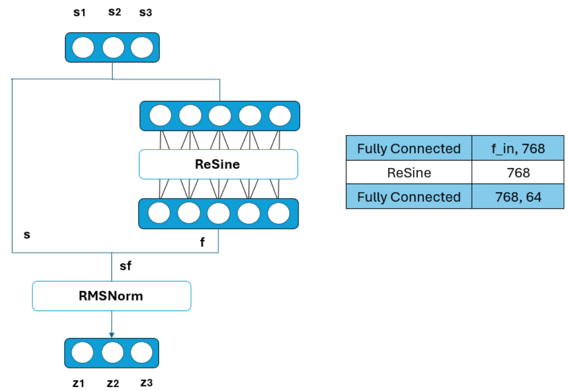



Fig. 11: Feature Extractor (shared).

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During the training of the Actor, Critic, and Reward Network, each network influences the gradient formation of the Feature Extractor, which outputs the latent state $z$. Unlike OFE or TD7, we do not utilize a separate network to process the concatenation of the latent state and actions ($za$).

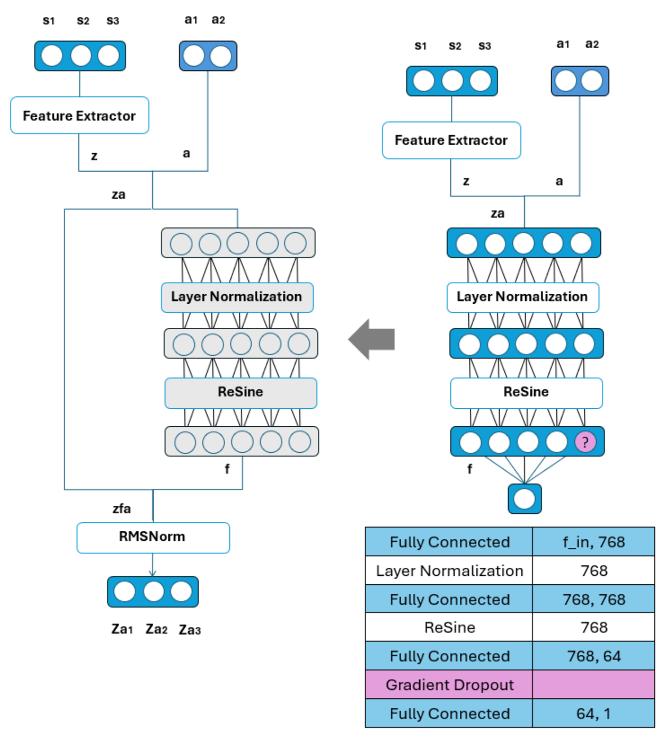


Fig. 12: Our simplified OFE architecture.

Instead, we employ a jointly trained shared Feature Network alongside the latent features of a non-trainable Reward Net-

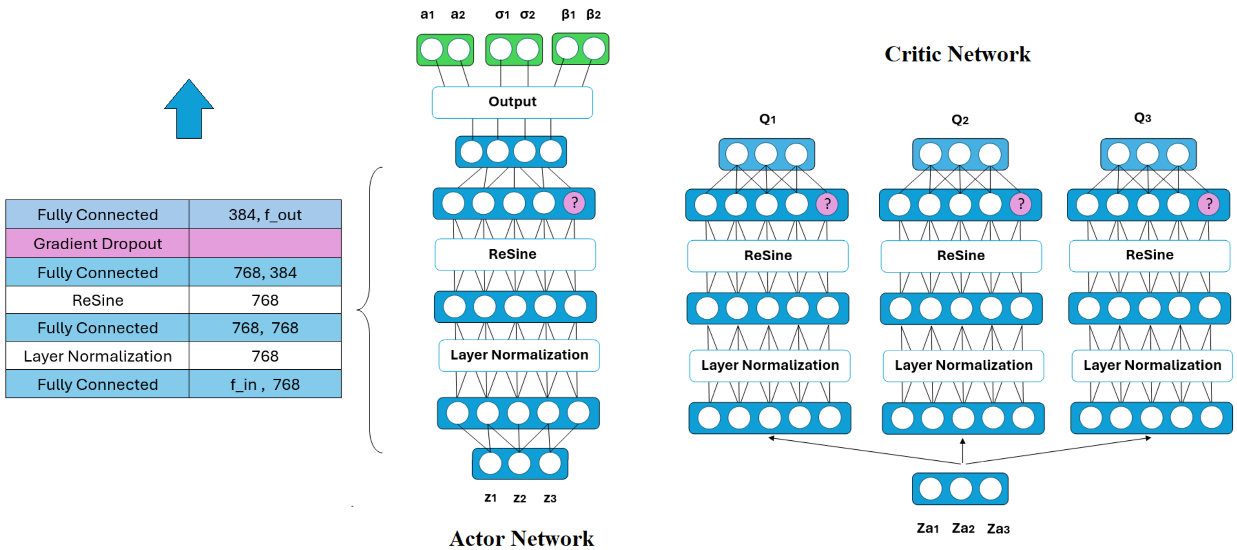


Fig. 10: Architecture of the Actor-Critic network

work (via a stop-gradient operator), while the Reward Network itself is trained on reward prediction. To normalize the output values, we use RMSNorm, which does not shift the mean to zero.

Gradient update of the Shared Feature Extractor happens simultaneously with seamless Actor-Critic and Reward Networks updates. The Reward Network is embedded inside Shared Feature Extractor which itself is part of the Actor-Critic object.

Predicting a normalized reward rather than the next state is a less complex task that captures the underlying valuable estimation of the current state-action pair.

All these hyperparameters and measures constitute the default implementation of the *Symphony* algorithm. Other configurations are possible, which result in different associated parameters. We do not recommend a Replay Buffer size lower than 7680*50=384,000.

*Noiseless conditions and auxiliary reward regularization*

For real-world motors, Gaussian noise can be dangerous. Consequently, for environments where Gaussian noise poses a operational risk, it must be entirely eliminated during environment interactions, with the sole exception of explicit exploration steps and network updates. In such cases, when actions are performed entirely without noise, the gradient may become too deterministic; therefore, we use parametric standard deviation of $1/2$ instead of $1/e$ during network updates.

Additionally, the following form of reward regularization using the next $\beta$ better prevents the Critic from prematurely converging and consequently, resists the Actor's aggressive pursuit of its objective, ultimately contributing to an overall improvement in performance.

$$\eta\,\widehat{r}_t + \gamma\left(q'^{*}_{t+1} - \mathcal{H}^{*}_{t+1}\right),\ \text{where}\ \eta = \frac{1}{d}\sum_{i=1}^{d}\beta_{t+1,i} \qquad (12)$$

This approach involves multiplying rewards by the mean of the $\beta$ distribution across its dimension. Rewards are scaled up by a higher state of $\beta$; meaning rewards with higher entropy receive higher estimates.

Although we rely on the averaged next $\beta$, the continuous nature of the neural network ensures that its value for the next state remains very close to the current $\beta$, while during training the average value of $\beta$ stabilizes within the range of 0.07 to 0.09, thereby effectively serving as a Q-value normalization factor.

## IV. Ablation Study

*Sample-efficient version, (SE)*

We made a Sample-efficient configuration by decreasing the noise level $\epsilon \sim a_{max} * 1/\pi * \mathcal{N}(0.0,\ 1.0)$, where $\mathcal{N} \in [-\pi, \pi]$.

*Sample-Proximity and Safety, (S2)*

It is possible to shift focus from Sample-Efficiency to Sample-Proximity and Safety just by increasing the noise level or sending $\beta$ with a higher order than 2.0 which will eventually punish strong action values harder. However, our Full Swaddling Function was balanced while the parametric noise value of 1/e perfectly suits it and already at the highest level. Layer Normalization and Resine Activation Function provide indirect generalization for the Input and Hidden Layers, respectively, whereas the Output Layer was left untouched. Original Dropout Layer [21] did not suit as it would zero some nodes distorting final distribution.

Last but not least we added a specific implementation of *Gradient Dropout* [22], which instead of zeroing some nodes as in Dropout Q Functions (DroQ) [23], does not calculate their gradients during backpropagation. This can be achieved simply by (mask is an original dropout mask, $x^*$ - detached values):

$$mask * x + (1 - mask) * x^* \tag{13}$$

Gradient Dropout after Fully Connected Output Layer in Actor and Critic with a dropout probability of 50% was the last missing ingredient. We, additionally, decreased the learning speed: $\alpha_{lr} = \frac{1}{2}\text{e}{-4}$ and increased initial generalization: $N_{exp} = 20,480$, $N_{rb} = 25\, N_{exp} = 512,000$.

*Embedded devices (ED), Baseline Model*

A configuration for less computationally powerful devices like Jetson Series: $N_{exp} = 7680$, $N_{rb} = 384,000$, $h_{dim} = 384$, each Critic outputs 96 nodes, $\mathcal{B} = 384$. Data is stored in Fading Replay Buffer as float16, converted to float32 during the sample retrieval process.

## V. Experiments

We set our experiments in Humanoid-v4, OpenAI's Gymnasium (Mujoco) Environment. Earlier versions of the Mujoco Humanoid environment, e.g. Humanoid-v1, had a maximum scale of action within -1 and 1, and due to the complexity of agent training, the maximum scale decreased to [-0.4, 0.4]. We returned the limit of the scale value to 1.0, since we have internal regularization. We did 10 random seed experiments for following models: Symphony-S3, Symphony-SE , Symphony-S2, Symphony-ED (Baseline). Symphony-S2 uses 20,480 exploratory steps, while others use 10,240. Total number of steps $3 * 10^6$, the episode limit was 1000 steps. We took readings in two ways, step-wise, when we sampled 25 trials each 2500 steps to estimate return and episode-wise to read an average $\sigma$ value in batch (Exploratory steps and episodes were omitted on the graphs and tables).

As was said for the Baseline model we choose Symphony-ED which to some extent close to Soft-Actor and Critic algorithm in performance but still generates alternating leg movements, though upper-body can stay underdeveloped, variance is relatively low:

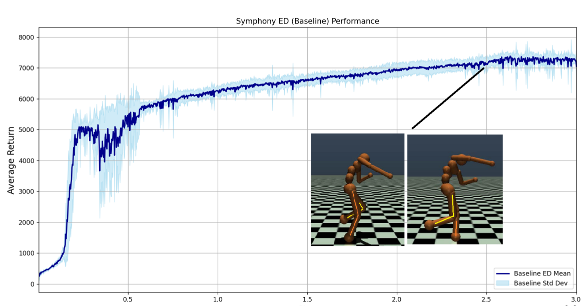


Fig. 13: Symphony-ED Return step-wise (orange lines depict supporting leg at a point in time).

In contrast, Sample-Efficient version, Symphony-SE develops a better gait, but suffer from high volatility especially when the agent's speed increases. It would be wise to decrease learning rate after some point.

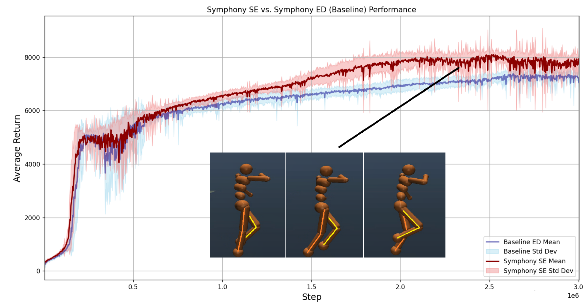


Fig. 14: Symphony-SE Return step-wise (orange lines depict supporting leg at a point in time)

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Symphony-S3 shows better development through the entire training, though as S3 is very close to SE model, it still suffers of higher variance at higher speeds. The same technique of decreasing learning rate can work here as well.

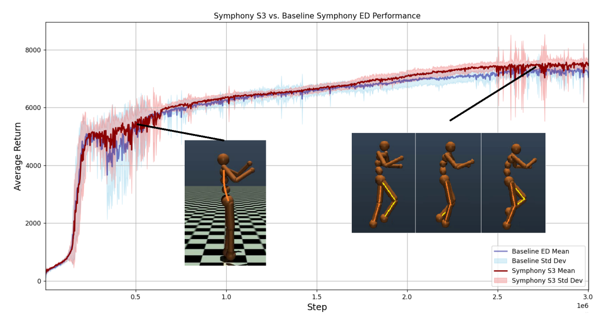


Fig. 15: Symphony-S3 Return step-wise (orange lines depict supporting leg at the moment)

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Symphony-S2 solves most problems of SE and S3 algorithms in exchange of sample-efficiency, however, we can see that the average Return shows positive dynamics, while for the Baseline model it might hit plateau.

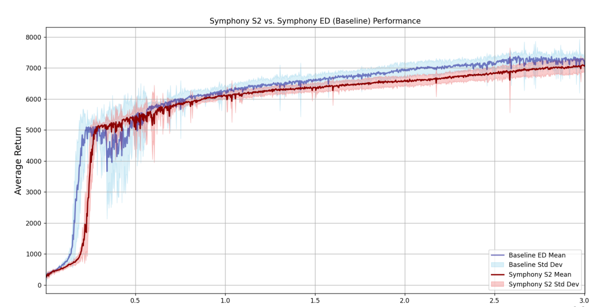


Fig. 16: Symphony-S2 Return step-wise

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The diagram below shows scaling factor behavior episode-wise. For each algorithm, we trimmed the values for the first episode when the number of steps reached $3 * 10^6$. It can be seen that the Scaling factor initially varies greatly (least of all for Symphony-S2), but then decreases closer to $\frac{1}{e}$, and slowly increases from this value.

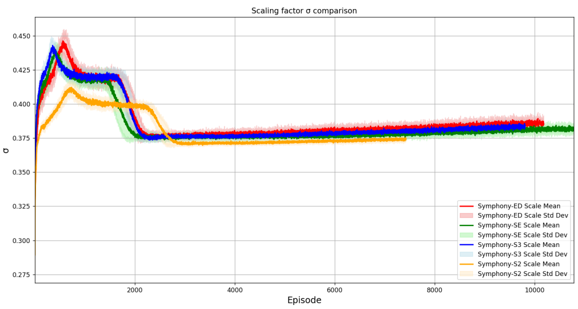


Fig. 17: Scaling Factor comparison.

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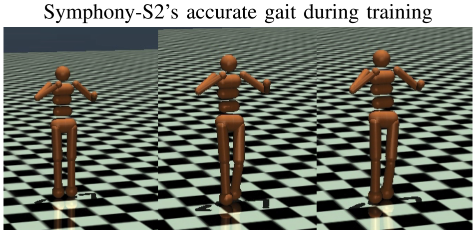
We were interested in finding out the average value for the episode when $3 * 10^6$ steps are reached. A preliminary conclusion can be drawn that, better generalization and slower learning rate result in better prediction as the agent falls less:

TABLE IV: Episode number when the $3 * 10^6$ steps were reached

| Version | Average episode |
|---|---|
| ED | $11530.5 \pm 1132.8$ |
| SE | $12433.5 \pm 1055.0$ |
| S3 | $11411.5 \pm 966.0$ |
| S2 | $\mathbf{8284.1 \pm 753.7}$ |

Symphony-S2's accurate gait during training



However, this does not imply that top scores are reached faster. To develop human-like movements one may need to sacrifice Sample-Efficiency to some extent.

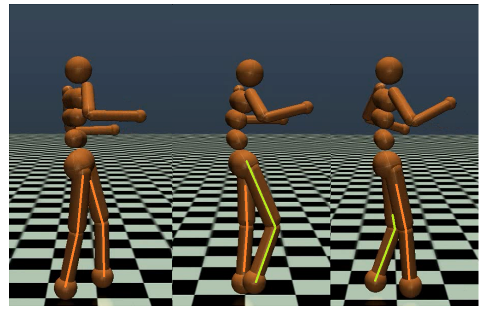


Fig. 18: Depiction when agent uses both hands and legs for balance during training

TABLE V: Step when the specific threshold of average Return is reached

| | The Symphony version | | | |
|---|---|---|---|---|
| Avg. Return | ED | SE | S3 | S2 |
| 5000 | 227k | **217k** | 240k | *285k* |
| 6000 | 782k | **657k** | 682k | *880k* |
| 7000 | 2,060k | **1,392k** | 1,895 | *2,768k* |

Even with all this, there's no $100\%$ guarantee that there will be absence of developmental defects. In our world, external obstacles often play an additional regulatory role. Following the logic of gradual human learning, we recommend starting training without obstacles, then gradually increasing the difficulty, filling a new Fading Replay Buffer with new exploratory data.

### *Symphony Saya (Sabbath)*

For practical use, we decided to stick with Symphony-S2, with some modifications.

At the simulation level, we utilized a limited bounded noise for the Actor Network, maintained consistently both within the training iterations and during environment interactions. However, for real-world motors, Gaussian noise can be dangerous. Consequently, for environments where Gaussian noise poses a operational risk, it must be entirely eliminated during environment interactions, with the sole exception of explicit exploration steps.

The surrogate approach involves multiplying rewards by the mean of the $\beta$ distribution across its dimension, while mean is found as well for $\Omega\omega$ and $\Omega$ functions across the action space. Rewards are scaled up by a higher state of $\beta$; meaning rewards with higher entropy receive higher estimates.

$$\eta * \hat{r}_t + \gamma\, Q^*_{\theta T}, \text{ where } \eta = \frac{1}{d}\sum_{i=1}^{d} \beta_{t+1,i} \tag{14}$$

Although this approach is an approximation due to the reliance on the averaged next $\beta$, the average value of $\beta$ stabilizes within the range of 0.07 to 0.09 during training, thereby effectively serving as a Q-value normalization factor while the continuous nature of the neural network ensures that its value for the next state remains very close to the current $\beta$. This surrogate approach better prevents the Actor from prematurely converging or aggressively pursuing its objective, ultimately contributing to an overall improvement in performance.

## Conclusion

In this work, we chose to prioritize stability over a fast convergence rate to eliminate jerky motions and prevent convergence to suboptimal policies during training in a single environment. In **Symphony Saya**, the update-to-data ratio was reduced to 1. Furthermore, we multiply the reward by a $\beta$ value that is unknown to the Critic, which introduces an additional challenge during the training process.. We acknowledge that our results might not achieve state-of-the-art benchmarks in any single direction as the Symphony algorithm is the result of rigorous work balancing three constraints - Sample Efficiency, Sample Proximity and Safety of Actions (regarding extreme action values) for practical applications. To describe the Symphony Algorithm in a nutshell, it can be considered as a biological organism where all parts complement each other: disabling one weakens another. Even though the update-to-data ratio is 3 it is mostly used for coherent understanding of the distribution shift. We conducted hundreds of experiments before finding suitable fixed parameters for humanoid robots. In the next article, we'd like to conduct a full analysis of each component separately, we are preparing an ablation study as a separate project.

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I want to thank my Lord, Jesus Christ. In the scientific community, this name, and religion in general, is not particularly welcome. I consider it important to say that I cannot even breathe without Him. He gives us life and the opportunity to create and build, He changes our character for the better. He gave me many ideas not when I spent hours on end sitting at the computer, but when I finally decided to spend time with my family, or when I listened to a professor. He teaches me to value, respect, and love my closest ones.

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